# DECOMPOSING THE SOURCES OF EARNINGS INEQUALITY:

# ASSESSING THE ROLE OF REALLOCATION

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Abstract

This paper uses matched employer-employee data from the U.S. Census Bureau to investigate the contribution of worker and firm reallocation to changes in wage inequality within and across industries between 1992 and 2003. We find that the entry and exit of firms and the sorting of workers and firms based on underlying worker skills are important sources of changes in earnings distributions over time. Our results suggest that the underlying dynamics driving changes in earnings inequality are complex and are due to factors that cannot be measured in standard cross-sectional data.

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## 1. Introduction

There has long been interest in disentangling the sources of changes in earnings inequality. However, while there is a large body of research on the evolution of the wage distribution, few studies have focused on the role of firm turnover in driving changes in earnings inequality, and fewer still have considered the importance of sorting among workers and firms in affecting the distribution of wages. This paper uses linked employer-employee data to examine whether the very high levels of job and worker reallocation in the U.S. economy, which have important implications for productivity, also affect the distribution of earnings within and across industries.

We develop a decomposition methodology that exploits our longitudinal data and permits us to quantify factors contributing to changes in earnings inequality over time. First, we re-examine the impact of changing workforce composition, particularly workforce skill and experience, which has been much studied with worker-based surveys. Second, we evaluate the impact of changing firm composition as firms expand, contract, enter, and exit in a given industry. Finally, we consider the impact on the earnings distribution of the way in which workers are sorted across firms. In addition to an examination of the overall economy, we control for the impact of changing industry structure by separately examining each of the nine major industries that comprise the private sector.

In general, we find that the pattern of earnings changes across industries is complex and heterogeneous. There is no single factor – workforce composition, firm composition, or the match between firms and workers - that can fully explain changes in the earnings distribution in the broad economy or in any given industry. Even when the direction of change is similar across industries, the underlying forces contributing to changes can be very different.

However, some common patterns are evident. Despite large amounts of worker turnover, entering workers are similar to exiting workers in most industries, and hence worker churning has little impact on wage inequality. In contrast, the entry and exit of firms have acted to reduce wage inequality, in particular by raising wages at the lower end of the wage distribution. Finally a higher degree of assortative matching of workers and firms has contributed to increasing wage inequality.

A major contribution of this paper is the finding that the extensive amount of ongoing worker and firm reallocation has large effects on different parts of the earnings distribution. In particular,

the entry and exit of firms and sorting of workers and firms based on underlying worker attributes are important sources of changes in industry earnings distributions over time. Changes in institutional factors, and in particular changes in the minimum wage, do not materially affect our main findings with respect to the critical role of firm turnover and worker sorting in giving rise to changes in wage inequality in recent years. However, we also find that changes in the composition of workers, the composition of firms, and the allocation of workers across firms cannot entirely account for changes in wage inequality, which is consistent with the view that other factors, such as changes in the returns to different skills, have played some role in shaping the distribution of earnings within and between industries.

The paper is organized as follows. After a brief review of the literature in Section 2, we develop an econometric method for decomposing the sources of change in the earnings distribution that takes advantage of employer-employee matched data in Section 3. We discuss the data we use in Section 4, and present some basic empirical facts about recent changes in earnings distributions in each of nine major industries in Section 5. Section 6 describes the results of performing the decompositions. We supplement the main results with an analysis of the implications of minimum wage legislation on observed patterns of earnings inequality in Section 7. Section 8 concludes.

## 2. Background

Despite a vast literature that attempts to explain the increase in earnings inequality that has occurred in recent decades, there is still not complete consensus on its primary sources (Levy and Murnane, 1992; Danziger and Gottschalk, 1995; Katz and Autor, 1999). A large body of research suggests that the increase in inequality was driven by skill-biased technical change interacting in complex ways with changes in unionisation, management structures, and international trade (Acemoglou, 2002). However, there is some disagreement about the relative importance of labour market versus institutional factors. For example, some researchers point to changes in the composition of the workforce as an important contributor to growing earnings inequality (Lemieux, 2006). Others, such as DiNardo et al. (1996), Lee (1999), and Card and DiNardo (2002), identify structural changes, such as the fall in the real value of the minimum wage and declines in unionisation, as the main drivers behind recent increases in inequality. Fortin and

Lemieux (1997) also suggest that deregulation in transportation, communication, and banking industries in the 1980s may have played a role.

In more recent work, Autor et al. (2008) present evidence that since the late 1980s, there has been a divergence in patterns of inequality between the upper and lower halves of the wage distribution, with the lower half (as measured by the 50-10 difference in log wages) either being compressed or not changing and the upper half (as measured by the 90-50 difference) exhibiting increasing dispersion. They argue that changes in the composition of the labour force have had a particularly pronounced effect on residual wage inequality increases in the upper half of the distribution, but less of an effect on the lower half.

Nearly all of the literature on wage inequality to date is based on surveys of workers, most notably the Current Population Survey (CPS). However, there is some firm-based evidence that suggests that changes in the distribution of wages may be due at least in part to changes on the firm side of the labour market. For example, there is research that indicates that patterns of firm entry and exit and the reallocation of resources across different sectors of the economy may be contributing to changes in earnings inequality. Bernard and Jensen (2000) find that increases in wage inequality across states are highly correlated with shifts in industrial composition, and in particular the decline in manufacturing. Burgess et al. (2001) also observe sizeable differences in trends in earnings inequality across industries in one U.S. state, while Davis and Haltiwanger (1991) show that the distribution of wages has changed over time even within the manufacturing sector.

The sheer magnitude of reallocation of workers across firms over time suggests that it could potentially have a very large impact on the earnings distribution. Davis et al. (1996) document the large and persistent rates of job creation and destruction in the U.S. economy and highlight the dominance of idiosyncratic factors in driving ongoing reallocation of workers across firms. Even within firms, the amount of worker turnover is large. Burgess et al. (2000) point out that after nine years, only about one third of private-sector workers are still employed by the same employer. Thus, as Burgess et al. (2009) also emphasize, there is ample scope for worker reallocation to affect earnings distributions.

The increasing availability of matched employer-employee data is providing new sets of facts that can inform our understanding of the impact of employers on earnings dispersion. First, it has been established that different firms pay observationally equivalent workers different wages.

Thus, variation in wage setting practices across firms can affect the earnings distribution. Second, within-firm analysis shows that very different wages are paid within each firm; in fact, the variation in earnings within firms is nearly as high as the variation in the overall earnings distribution (Lane, 2009). Therefore, changes in compensation policies within firms can affect wage inequality. Finally, there is a burgeoning literature that suggests that the way in which workers are matched to different types of firms is non-random; in particular, high wage workers tend to be matched with high wage firms and low wage workers with low wage firms. Changes in that allocation can change earnings distributions over time (Abowd et al., 1999; Lane, 2009).

Taking advantage of longitudinal matched employee-employer data, we focus in this study on the impact of not only changes in the types of workers and the types of firms in different industries, but also changes in the allocation of workers across firms within industries on changes in earnings distributions over time. We turn to a discussion of how we decompose changes in earnings inequality using panel data on workers and firms in the next section.

## 3. Decomposition Methodology

In this section, we develop an approach to decomposing changes in earnings distributions that exploits employer-employee matched panel data. The aim is to decompose the change in earnings inequality observed between two time periods into portions attributable to changes in the types of workers, changes in the types of firms, and changes in the allocation of workers across firms.

For simplicity, we initially assume that we have only one continuous exogenous predictor variable, $x$. Let $\theta$ be a variable representing an individual's (unobserved) productivity that is assumed to be constant over time. Further, let $\psi$ represent a firm's (unobserved) pay policy variable that is also assumed to be constant over time. An individual's earnings are assumed to be determined by the function $y = g(\varepsilon, \theta, \psi, x)$, where $\varepsilon$ is a random error component that is assumed to be independent of $x$, $\theta$, and $\psi$ that will capture a number of factors discussed below, including changes in the returns to observable and unobservable characteristics. We assume that the variables $x$, $\theta$, $\psi$, and $\varepsilon$ have a continuous joint probability density function $f_t$ for each time period $t = 1, 2$. Within an industry, firms can be created or destroyed and workers may enter or exit over time. Thus, for both workers and firms, there are stayers ($s$), leavers ($l$),

and new entrants (*n*). We can therefore write the joint density at time period 1 as a mixture of these worker-firm types,

$$(1)\quad \begin{aligned} f_1(\varepsilon,\theta,\psi,x) = {} & p_1(\text{w=}s,\ \text{f=}s) f_1^{ss}(\varepsilon,\theta,\psi,x) + p_1(\text{w=}l,\ \text{f=}s) f_1^{ls}(\varepsilon,\theta,\psi,x) + \\ & p_1(\text{w=}s,\ \text{f=}l) f_1^{sl}(\varepsilon,\theta,\psi,x) + p_1(\text{w=}l,\ \text{f=}l) f_1^{ll}(\varepsilon,\theta,\psi,x) \end{aligned},$$

where $p_1$(w=s, f=*s*) is the fraction of worker-firm matches where both worker and firm remain in the industry until time 2, $p_1$(w=*l*, f=*s*) is the fraction of worker-firm matches where the worker leaves the industry between time 1 and time 2 but the firm remains, $p_1$(w=*s*, f=*l*) is the fraction of worker-firm matches where the worker remains in the industry until time 2 but the firm leaves, and $p_1$(w=*l*, f=*l*) is the fraction of worker-firm matches where both the worker and firm leave by time 2. $f_1^{ss}(\varepsilon,\theta,\psi,x)$, $f_1^{ls}(\varepsilon,\theta,\psi,x)$, $f_1^{sl}(\varepsilon,\theta,\psi,x)$, and $f_1^{ll}(\varepsilon,\theta,\psi,x)$ are the corresponding conditional densities.

Similarly, the joint density for time period 2 as can be written as

$$(2)\quad \begin{aligned} f_2(\varepsilon,\theta,\psi,x) = {} & p_2(\text{w=}s,\ \text{f=}s) f_2^{ss}(\varepsilon,\theta,\psi,x) + p_2(\text{w=}n,\ \text{f=}s) f_2^{ns}(\varepsilon,\theta,\psi,x) + \\ & p_2(\text{w=}s,\ \text{f=}n) f_2^{sn}(\varepsilon,\theta,\psi,x) + p_2(\text{w=}n,\ \text{f=}n) f_2^{nn}(\varepsilon,\theta,\psi,x) \end{aligned},$$

where *n* indicates new entrants into the industry (whether they be workers or firms) between time 1 and time 2.

Results from the Oaxaca-style decomposition that follows can be sensitive to the order of variables. Several different orderings yielded similar results, and here we present the case in which we first analyze the extent to which worker entry and exit changed the earnings distribution between period 1 and 2. Had there been no worker entry and exit, (2) becomes

$$(3)\quad f_2^{w}(\varepsilon,\theta,\psi,x) = \frac{p_2(\text{w=}s,\ \text{f=}s) f_2^{ss}(\varepsilon,\theta,\psi,x) + p_1(\text{w=}l,\ \text{f=}s) f_1^{ls}(\varepsilon,\theta,\psi,x) + p_2(\text{w=}s,\ \text{f=}n) f_2^{sn}(\varepsilon,\theta,\psi,x)}{R}$$

where $R = p_2(\text{w=}s,\ \text{f=}s) + p_1(\text{w=}l,\ \text{f=}s) + p_2(\text{w=}s,\ \text{f=}n)$. Here, we have assumed that, had those individuals who left instead stayed, they would have matched with firms in a manner analogous to the distribution of workers who actually left those firms that stayed in the industry.

Next, we consider the impact of the change in the distribution of observable worker characteristics, $x$. Here, we note that, for example, $f_2^{ss}(\varepsilon,\theta,\psi,x) \equiv f_2^{ss}(\varepsilon,\theta,\psi \mid x) f_2(x)$, and replace $f_2(x)$ by $f_1(x)$ to give us

$$(4)\quad f_2^{ss,x}(\varepsilon,\theta,\psi,x) \equiv f_2^{ss}(\varepsilon,\theta,\psi \mid x) f_1(x) = f_2^{ss}(\varepsilon,\theta,\psi,x)\left( f_1(x) \Big/ f_2(x) \right).$$

The other terms in (3) are modified in a similar fashion. Thus, we have

$$(5)\quad f_2^{w,x}(\varepsilon,\theta,\psi,x) = \frac{p_2(\text{w}=s,\ \text{f}=s) f_2^{ss,x}(\varepsilon,\theta,\psi,x) + p_1(\text{w}=l,\ \text{f}=s) f_1^{ls,x}(\varepsilon,\theta,\psi,x) + p_2(\text{w}=s,\ \text{f}=n) f_2^{sn,x}(\varepsilon,\theta,\psi,x)}{R}.$$

Next, we evaluate the impact of firm entry and exit by considering the counterfactual that assumes that the set of firms (as well as workers and $x$) at time 2 is the same as time 1:

$$(6)\quad f_2^{w,x,e}(\varepsilon,\theta,\psi,x) = \frac{p_1(\text{w}=s,\ \text{f}=s) f_2^{ss,x}(\varepsilon,\theta,\psi,x) + p_1(\text{w}=l,\ \text{f}=s) f_1^{ls,x}(\varepsilon,\theta,\psi,x) + p_1(\text{w}=l,\ \text{f}=l) f_1^{ll,x}(\varepsilon,\theta,\psi,x)}{R'}$$

where $R' = p_1(\text{w}=s,\ \text{f}=s) + p_1(\text{w}=l,\ \text{f}=s) + p_1(\text{w}=l,\ \text{f}=l)$.

Finally, after we have restricted the set of firms and workers to be the same as in period 1, it is still possible to examine how the conditional distribution of $\theta$ given $\psi$ may have changed between periods 1 and 2 due to a reallocation of workers across firms within the industry.[1] Now,

$$(7)\quad \begin{aligned} f_2^{w,x,e}(\varepsilon,\theta,\psi,x) &\equiv f_2^{w,x,e}(\varepsilon,\theta,\psi,x \mid \theta,\psi) f_2^{w,x,e}(\theta,\psi) \\ &= f_2^{w,x,e}(\varepsilon,\theta,\psi,x \mid \theta,\psi) f_2^{w,x,e}(\theta \mid \psi) f_1(\psi) \end{aligned}.$$

Hence, we can define

[1] Note that at this point in the decomposition, the marginal distributions of $\theta$ and $\psi$ are those of period 1 since the set of firms and workers are the same as period 1. Thus, only the conditional distribution $f_2^{w,x,e}(\theta \mid \psi)$ (or, alternatively, $f_2^{w,x,e}(\psi \mid \theta)$) can differ.

$$(8) \qquad f_2^{w,x,e,a}(\varepsilon,\theta,\psi,x) \equiv f_2^{w,x,e}(\varepsilon,\theta,\psi,x \mid \theta,\psi) f_1^{w,x,e}(\theta \mid \psi) f_1(\psi),$$

where the superscript refers to holding the allocation of workers to firms constant.

From these counterfactual densities, we can decompose changes in the earnings distribution. Let $Y$ be the range of $y$ and let $A$ be a subset of $Y$ (i.e., $A \subset Y$ ). Then, the probability that $y$ falls within $A$, $P_t(y \in \mathrm{A})$, is

$$(9) \qquad P_t(y \in A) = \int_{\{\varepsilon,\theta,\psi,x\}: g(\varepsilon,\theta,\psi,x) \in A} f_t(\varepsilon,\theta,\psi,x) d\varepsilon d\theta d\psi dx .$$

We can then define the counterfactual probabilities by

$$(10) \qquad P_2^{w}(y \in A) = \int_{\{\varepsilon,\theta,\psi,x\}: g(\varepsilon,\theta,\psi,x) \in A} f_2^{w}(\varepsilon,\theta,\psi,x) d\varepsilon d\theta d\psi dx ,$$

$$(11) \qquad P_2^{w,x}(y \in A) = \int_{\{\varepsilon,\theta,\psi,x\}: g(\varepsilon,\theta,\psi,x) \in A} f_2^{w,x}(\varepsilon,\theta,\psi,x) d\varepsilon d\theta d\psi dx ,$$

$$(12) \qquad P_2^{w,x,e}(y \in A) = \int_{\{\varepsilon,\theta,\psi,x\}: g(\varepsilon,\theta,\psi,x) \in A} f_2^{w,x,e}(\varepsilon,\theta,\psi,x) d\varepsilon d\theta d\psi dx ,$$

and

$$(13) \qquad P_2^{w,x,e,a}(y \in A) = \int_{\{\varepsilon,\theta,\psi,x\}: g(\varepsilon,\theta,\psi,x) \in A} f_2^{w,x,e,a}(\varepsilon,\theta,\psi,x) d\varepsilon d\theta d\psi dx .$$

The Oaxaca-style decomposition of the change in the probability of the event $y \in A$ can then be written as

$$\begin{aligned} (14) \qquad & P_2(y \in A) - P_1(y \in A) = \\ & (P_2(y \in A) - P_2^{w}(y \in A)) + (P_2^{w}(y \in A) - P_2^{w,x}(y \in A)) \\ & + (P_2^{w,x}(y \in A) - P_2^{w,x,e}(y \in A)) + (P_2^{w,x,e}(y \in A) - P_2^{w,x,e,a}(y \in A)) \\ & + (P_2^{w,x,e,a}(y \in A) - P_2(y \in A)) \end{aligned} .$$

Therefore, supposing that we wish to decompose the expected value of some function *r* of earnings, *E(r(y))*, we have

$$
\begin{aligned}
&E_2(r(y)) - E_1(r(y)) = \\
&\int_{\{\varepsilon,\theta,\psi,x\}\in D} r(g(\varepsilon,\theta,\psi,x)) f_2(\varepsilon,\theta,\psi,x) d\varepsilon d\theta d\psi dx - \int_{\{\varepsilon,\theta,\psi,x\}\in D} r(g(\varepsilon,\theta,\psi,x)) f_1(\varepsilon,\theta,\psi,x) d\varepsilon d\theta d\psi dx = \\
&\left( \int_{\{\varepsilon,\theta,\psi,x\}\in D} r(g(\varepsilon,\theta,\psi,x)) f_2(\varepsilon,\theta,\psi,x) d\varepsilon d\theta d\psi dx - \int_{\{\varepsilon,\theta,\psi,x\}\in D} r(g(\varepsilon,\theta,\psi,x)) f_2^{w}(\varepsilon,\theta,\psi,x) d\varepsilon d\theta d\psi dx \right) + \\
&\left( \int_{\{\varepsilon,\theta,\psi,x\}\in D} r(g(\varepsilon,\theta,\psi,x)) f_2^{w}(\varepsilon,\theta,\psi,x) d\varepsilon d\theta d\psi dx - \int_{\{\varepsilon,\theta,\psi,x\}\in D} r(g(\varepsilon,\theta,\psi,x)) f_2^{w,x}(\varepsilon,\theta,\psi,x) d\varepsilon d\theta d\psi dx \right) + \\
&\left( \int_{\{\varepsilon,\theta,\psi,x\}\in D} r(g(\varepsilon,\theta,\psi,x)) f_2^{w,x}(\varepsilon,\theta,\psi,x) d\varepsilon d\theta d\psi dx - \int_{\{\varepsilon,\theta,\psi,x\}\in D} r(g(\varepsilon,\theta,\psi,x)) f_2^{w,x,e}(\varepsilon,\theta,\psi,x) d\varepsilon d\theta d\psi dx \right) + \\
&\left( \int_{\{\varepsilon,\theta,\psi,x\}\in D} r(g(\varepsilon,\theta,\psi,s)) f_2^{w,x,e}(\varepsilon,\theta,\psi,x) d\varepsilon d\theta d\psi dx - \int_{\{\varepsilon,\theta,\psi,x\}\in D} r(g(\varepsilon,\theta,\psi,x)) f_2^{w,x,e,a}(\varepsilon,\theta,\psi,x) d\varepsilon d\theta d\psi dx \right) + \\
&\left( \int_{\{\varepsilon,\theta,\psi,x\}\in D} r(g(\varepsilon,\theta,\psi,s)) f_2^{w,x,e,a}(\varepsilon,\theta,\psi,x) d\varepsilon d\theta d\psi dx - \int_{\{\varepsilon,\theta,\psi,x\}\in D} r(g(\varepsilon,\theta,\psi,x)) f_1(\varepsilon,\theta,\psi,x) d\varepsilon d\theta d\psi dx \right)
\end{aligned}
$$

where *D* denotes the domain of $(\varepsilon,\theta,\psi,x)$. Note that in (14) we have $r(y) = I(y \in A)$, where *I*(.) is the indicator function.[2] Thus, we can decompose the overall change into those parts attributable to worker entry and exit (the first term in parentheses), changes in observable worker characteristics controlling for worker entry and exit (the second term), firm entry and exit controlling for changes in workforce composition (the third term), changes in the allocation of workers across firms controlling for workforce and firm composition (the fourth term), and a residual (the fifth term).

The residual component may capture several possible effects. First, it will reflect any systematic changes in the returns to observable or unobservable characteristics between periods 1 and 2 since the decomposition as specified assumes that these returns are constant. Importantly, while our decomposition is designed to quantify the importance of turnover and sorting in bringing about changes in earnings distributions over time, we do not explicitly address the issue

[2] This decomposition technique can be extended to the case where the earnings function $y = g(\varepsilon,\theta,\psi,x)$ varies across time (i.e., $y = g_t(\varepsilon,\theta,\psi,x)$) by incorporating an additional decomposition step that would measure the impact of this "structural" change on the distribution of earnings. As our current focus is largely on the role of reallocation in explaining observed changes in the earnings distribution, we leave this extension for future work.

of changes in the returns to skills in the decomposition. To the extent that the returns to observable and unobservable characteristics change systematically over the time period we consider in our application, it will be captured in the residual. Changes in the "quality" of worker-firm matches that are not due to merely the reallocation of different workers across different firms will also show up in the residual component. Also, it will potentially capture changes in institutional factors that affect the earnings distribution. We consider one such factor, minimum wage legislation, in Section 7.

We apply this general decomposition technique to employee-employer matched panel data in order to explore the sources of changes in earnings distributions over time for different industries. To put more structure on the relationship between $y$ and $(\varepsilon, \theta, \psi, x)$, we assume that the relationship takes the form of a linear panel data model with fixed firm and individual effects, as in Abowd et al. (1999). That is, we assume that the function $g$ has the following form:

$$(15) \qquad y_{it} = g(\varepsilon, \theta, \psi, \mathbf{x}) = \mathbf{x_{it}}\boldsymbol{\beta} + \theta_i + \psi_{j(i,t)} + \varepsilon_{it} \ ,$$

where $y_{it}$ are individual $i$'s earnings at time $t$, $\mathbf{x}_{\text{it}}$ is a vector of observed time-varying personal characteristics of individual $i$ at time $t$, $\theta_i$ is an individual fixed effect, and $\psi_{j(i,t)}$ the fixed effect of the firm $j$ for which individual $i$ works at time $t$.

In essence, the individual fixed effect $\theta_i$ captures the portable component of an individual's earnings, or that component that belongs to an individual as he or she moves from job to job and that is separate from the type of firm for which he or she works. This person effect, which represents all non-time varying observable and unobservable individual heterogeneity, is our measure of human capital. In interpreting the human capital measure, several remarks should be made. First, the human capital measure is not simply a ranking of the earnings of the worker, precisely because earnings include both person and firm effects. Second, the person effect will reflect the influence of any time-invariant personal characteristics. Thus, for each individual, it will reflect factors including educational attainment, other observable accumulated skill correlates, and unobserved dimensions of skill. At the same time, it abstracts from firm-specific factors that may be present in measures based upon observable characteristics.

The firm effect $\psi_{j(i,t)}$, meanwhile, captures the extent to which the firm at which a worker is employed pays above or below average earnings (after controlling for person effects). The firm effect may reflect many factors including capital intensity, rent sharing, firm-specific human capital, compensating differentials, or unionisation effects (Abowd et al., 2003; Andersson et al., 2005). Changes in the distribution of firms that pay relatively high and low wages over time across sectors due to entry and exit could also contribute to changes in earnings inequality. Moreover, changes in the joint distribution of worker and firm effects, or changes in the extent to which high wage workers match with high wage firms, could affect the distribution of earnings (Abowd et al., 1999; Lane, 2009).

With a large number of observations, the approach outlined above can become impracticable. To circumvent this problem, one can estimate the continuous distribution of $(\varepsilon, \theta, \psi)$ for several categories of **x**. We discretized each variable by breaking the range into 100 mutually exclusive intervals and assigning the midpoint value to each observation that falls within the interval. This method is applied for all intervals except the lowest and highest intervals (which are unbounded). For the highest (lowest) interval, we assign a value that equals the average of the lower (higher) boundary value and the highest (lowest) observed value in the (industry) sample. We denote the discretized vales by $(\varepsilon^d, \theta^d, \psi^d)$. Earnings are then recomputed using the discretized values by

$$y_{it}^d = \mathbf{x_{it}}\boldsymbol{\beta} + \theta_i^d + \psi_{j(i,t)}^d + \varepsilon_{it}^d \,. \tag{16}$$

The decompositions are then performed on $y_{it}^d$ using the discrete analogs of the equations presented above.

## 4. Data

To decompose the sources of change in earnings inequality within industries, we take advantage of a database created by the Longitudinal Employer and Household Dynamics (LEHD) Program at the U.S. Census Bureau. These confidential data enable us to match workers with past and present employers, together with employer and worker characteristics (Abowd et al., 2004). This database consists of quarterly records of the employment and earnings of almost

all individuals from the unemployment insurance (UI) systems of a number of U.S. states in the 1990s and 2000s.

These data have been extensively described elsewhere (Haltiwanger et al., 2006), but it is worth noting several advantages of the LEHD data. The data are current, and the dataset is extremely large. Since the scope of the data is almost the universe of employers and workers in the covered private sector, it is possible to follow workers as they move between employers and along the earnings distribution.[3] The UI records have also been matched to internal administrative and survey data containing information on date of birth, place of birth, race, and sex for all workers, thus providing limited demographic information.

Of particular importance given the focus of this study is the highly accurate reporting of both earnings and industry in the LEHD data. Hirsch and Schumacher (2004) point out that as many of 30% of respondents to the CPS, the major source of information on earnings inequality in the literature, do not respond to income questions. As a result, their answers are imputed. In the LEHD data, earnings are quite accurately and universally reported by firms due to financial penalties for misreporting or failing to report. In addition, research comparing earnings and employer characteristics as reported by survey respondents to those recorded in administrative files suggests that workers not only often misreport earnings, but also frequently fail to identify their industry correctly even at the major industry level (Decressin et al., 2006).

The LEHD data have several limitations. First, the data are not available for all states, and the amount of historical data varies by state. As such, we isolate attention to four large states (California, Illinois, Maryland, and North Carolina) for which we have data for the period 1992-2003. Based on data from the CPS, these four states accounted for approximately one-fifth of total U.S. employment in 2003 and, taken together, are similar in industrial composition as the nation as a whole.

A second limitation of the LEHD data is that they lack information on hours worked, which makes it impossible to calculate an hourly wage rate or determine full-time or part-time status. Therefore, our measure of earnings is real (2003 dollars) annualized earnings, which is the full-year equivalent of the hourly real wage. To calculate this measure, we use only earnings from each worker's dominant employer, or that employer that contributes the most to the worker's total earnings in each year. To eliminate workers with minimal attachment to the labour market

[3] See Stevens (2002) for a description of UI data coverage.

and those employed only part of a quarter (and hence whose reported earnings represent compensation for an indeterminate amount of time), we also use only workers who have real earnings of at least $250 in at least one quarter of the year and who are full-quarter employed, where being full-quarter employed in quarter t is defined as having an employment history with positive earnings for quarters $t$ - 1, $t$, and $t$ + 1. Abowd et al. (2003) contains further details on constructing samples based on LEHD data.

Using these employee-employer matched data, we can derive unique measures of human capital and firm pay policies. In particular, based on equation (16) above, we can estimate individual and firm effects, $\theta_i$ and $\psi_{j(i,t)}$, controlling for time-varying personal characteristics $\mathbf{x}_{it}$. Though we focus on only two years, 1992 and 2003, in our decomposition, we estimate equation (16) based on annual data over the entire period over which we have historical data on each state.[4] In our application, $\mathbf{x}_{it}$ includes a quadratic in labour force experience and a set of work history dummies (to capture censored employment spells), all interacted with gender. With estimates of $\mathbf{x}_{it}\boldsymbol{\beta}$, $\theta_i$, and $\psi_{j(i,t)}$ in hand, we can in turn quantify the importance of changes in workforce composition (both due to changes in observable characteristics among stayers and due to worker entry and exit), changes firm composition, and changes in the allocation of different workers across different firms to changes in the overall distribution of earnings. The estimated residual is also informative regarding the importance of other potential factors that might drive changes in earnings inequality, such as changes in the returns to skills or minimum wage legislation.

While previous studies have made use of observable measures of human capital such as education and experience, rarely have researchers been able to control adequately for unobservable productive characteristics such as ability or interpersonal skills. Changes in unobservable characteristics of workforces across industries as well as changes in how workers with different levels of unobservable skills are allocated across different types of firms within industries could help to explain changes in earnings inequality. Similarly, controlling for unobservable firm characteristics that might affect compensation structures has been difficult in the past given data limitations. The LEHD dataset allows us to quantify the contributions of such

[4] Data used in estimating equation (16) span 1992-2003 for California, 1991-2003 for Illinois, 1986-2003 for Maryland, and 1991-2003 for North Carolina.

unobservable characteristics to earnings, although it does not permit a decomposition of their sources (Abowd et al., 1999; Abowd et al., 2003).

For the purposes of our analysis, we restrict attention to all private establishments and use the Standard Industrial Classification (SIC) to identify industry sectors. In particular, we focus on agriculture, forestry, and fishing (which we henceforth call simply agriculture) (SICs 01-09); mining (SICs 10-14); construction (SICs 15-17); manufacturing (SICs 20-39); transportation, communication, and utilities (TCU) (SICs 40-49); wholesale trade (SICs 50-51); retail trade (SICs 52-59); finance, insurance, and real estate (FIRE) (SICs 60-67); and services (SICs 70-89). We exclude public administration due to incomplete data coverage in that sector.

## 5. Basic Facts

In this section, we present basic descriptive statistics regarding overall wage inequality, workforce composition, firm turnover, and the allocation of workers across firms within and between industries in our sample. In the subsequent section, we turn to our decomposition results, which shed light on the relative importance of different factors in explaining changes in earnings distributions over time.

### *5.1 Changes in inequality*

We first consider basic characteristics of the earnings distribution in each of the nine major industries and all industries combined by calculating levels and log differences in real annualized earnings at different percentiles. Table 1 shows the $90^{th}$, $50^{th}$ (median), and $10^{th}$ percentile of earnings in 2003 and the 90-10, 90-50, and 50-10 log earnings differences by sector and for all sectors together.

An examination of the first three columns of Table 1 reveals that there are substantial earnings differences across industries. For example, median earnings are over twice as high in mining as in the agricultural sector, and similar differences hold at both the $90^{th}$ and $10^{th}$ percentiles. Earnings at the high and low end of the distribution also vary greatly across sectors. The highest $90^{th}$ percentile earnings are found in FIRE ($114,428), while the lowest $10^{th}$ percentile earnings are in retail and agriculture (both under $10,000). The distribution of

earnings also varies across sectors, particularly the 90-10 and 90-50 log earnings differences. The 2003 90-10 log earnings gap is highest in services, followed by FIRE, wholesale trade, retail trade, and manufacturing. These same five industries also had the highest 90-50 log earnings differences in 2003. In contrast, inequality at the lower end of the earnings distribution does not vary as much across industries, though services had the largest 50-10 log earnings difference.

As the last row of Table 1 shows, the 90-10 earnings gap increased by 0.06 log points between 1992 and 2003 across all sectors in the sample. Looking across industries, though, earnings inequality as measured by the 90-10 log earnings difference declined in four industries (agriculture, mining, construction, and retail trade) and increased in the remaining five industries (manufacturing, TCU, wholesale trade, FIRE, and services). The latter five industries accounted for about three-fourths of total employment in the sample each year, with services alone accounting for roughly one-third of total employment. The most marked increases in inequality, though, were in FIRE, manufacturing, and TCU.

In order to compare changes in earnings inequality in the upper and lower tails of the earnings distribution, Table 1 also breaks out changes in the 90-50 and 50-10 log earnings gaps between 1992 and 2003 by industry. There are clear differences in trends in inequality across sectors. In three of the four industries in which overall inequality (the 90-10 log earnings difference) declined, much if not all of the decrease was in the lower half of the earnings distribution (the 50-10 log earnings gap). Only in mining was there much of a decline in the upper half (the 90-50 log difference). In contrast, earnings inequality in the lower end of the distribution did not increase in the five industries in which overall inequality increased. The increase in the 90-10 log earnings difference in manufacturing, TCU, wholesale trade, FIRE, and services was driven almost entirely by an increase in the spread between the $90^{th}$ and the $50^{th}$ percentiles of earnings.

Overall, the results reported in Table 1 are consistent with Autor et al. (2008), who find using CPS data that economy-wide, the 90-50 earnings gap grew during the 1990s while the 50-10 difference leveled off after about 1987. However, we also see differences in trends in upper and lower tail inequality across sectors. This heterogeneity suggests that different labour market or institutional factors may have affected different industries in different ways. In what follows, we discuss some of these factors and later attempt to identify the relative importance of each.

*5.2 Workforce composition*

One possible reason for changes in the earnings distribution in any given industry is that the workforce composition has changed over time. Table 2 suggests that there is ample potential for such changes to occur. In manufacturing, for example, of the more than five million workers who were employed in 1992, 2003, or both years, 44% were only in the industry in 1992, 35% were only in the industry in 2003, and 21% were there (but not necessarily in the same firm) in both years. As one might expect, turnover in the workforce is even more marked in the wholesale and retail trade industries, where only 12% of workers in each industry in 1992 were still in the same industry in 2003.[5]

The substantial turnover within industries was accompanied by dramatic changes in the distribution of workers across industries, as Table 3 shows. While the mining and manufacturing industries experienced double-digit drops in employment in percentage terms between 1992 and 2003, construction and services industries witnessed over 40% increases in employment over the same period. However, these structural changes did not coincide with enormous swings in the age, gender, and skill distribution of workers within industries. In other words, even as some industries shrank and others expanded, the workforce characteristics of each changed very little. Mining and manufacturing, for example, remained over two thirds male and skewed toward older workers. In contrast, industries such as FIRE and services continued to employ more females and younger workers. Similarly, although the average skill level of the workforce increased in all industries (using the individual fixed effect as a measure of skill), the swings are not substantial.

*5.3 Firm turnover*

Another possible reason for changes in earnings inequality is changes in the types of firms that are hiring workers. We examine this possibility in Table 4, which can be read the same way as Table 2. For example, in manufacturing, of the more than 100,000 firms that employed

[5] Workers can enter a particular industry either by leaving another industry or by entering the sample over the period. Similarly, workers can exit an industry by moving to another industry or leaving the sample. Workers entering the sample can be new labour force entrants who resided in our group of states or migrants from other states. Workers leaving the sample can be workers leaving the labour force or workers moving to a state outside our group of states. Unfortunately, we can not distinguish these different groups of sample entrants and leavers.

individuals in 1992, 2003, or both years, about 36% were only in the industry in 1992, 37% were only in the industry in 2003, and only 27% were there in both years. Survival rates are even lower in industries with more small firms. For instance, in retail trade and services, only about 20% appear in both years. These high levels of firm turnover are consistent with past research; Faberman and Freedman (2009) and others have also found that over half of new establishments do not survive more than five years, and that the highest turnover rates are in retail and services. To the extent that entering, exiting, and continuing firms vary in their characteristics and compensation policies, firm turnover could also lead to changes in earnings inequality over time.

*5.4 Sorting*

Another potential source of change in earnings inequality is changes in the joint distribution of worker and firm characteristics. In other words, changes in the allocation of different workers across different firms could affect the distribution of earnings.

In terms of the two-way fixed effects model described in Section 3, changes in the distribution of earnings may be due to changes not only in $\theta$ and $\psi$ independently, but also in the joint distribution of $\theta$ and $\psi$. For example, over time it may be the case that high $\theta$ individuals are more likely to work at high $\psi$ firms and that low $\theta$ individuals are more likely to work at low $\psi$ firms (Abowd et al., 1999). All else equal, this trend would tend to increase earnings inequality over time.

Using predicted values of $\theta$ and $\psi$ from the model, Figure 1 plots the joint distributions of $\theta$ and $\psi$ for 1992 and 2003. As the mass points in the left and right corners of each figure suggest, there is some tendency for low $\theta$ workers to be employed with low $\psi$ firms and for high $\theta$ workers to be employed with high $\psi$ firms. While this pattern held in both 1992 and 2003, a closer look at the extremes of the joint distribution of individual and firm effects reveal an interesting trend suggestive of greater assortative matching. The expected average $\theta$ (deviated from year means) more than tripled from 0.75 to 2.50 between 1992 and 2003 for the top 5% of firms as ranked by their $\psi$ s. Meanwhile, the expected average $\theta$ remained roughly constant for the bottom 5% of firms at -0.75. Thus, individuals with very high skill levels were more likely in 2003 than 1992 to be paired with firms with high pay policies, while those with

low skill levels were not. Whether this finding is a result of entry and exit of different types of firms and workers or is due to a reshuffling of workers across different firms, however, requires further investigation. In the next section, we apply our decomposition approach to examine different possible sources of changes in inequality and to quantify their relative importance.

## 6. Decomposition Results

### *6.1 Results for all industries*

Using the LEHD data, we first decompose the change in the earnings distribution for all nine sectors combined between 1992 and 2003. In decomposing changes in the earnings distribution for all sectors, we must account for the change in the employment distribution across sectors over time; the contribution of such changes is considered first in the decomposition. That is followed by estimates of the contributions of worker entry and exit, changes in observable worker characteristics, firm entry and exit, and changes in the distribution of worker unobserved attributes ($\theta$) for a given firm pay policy ($\psi$) – i.e., sorting. That leaves us with a residual component that could reflect factors not considered explicitly in the decomposition, including changes in the returns to characteristics.

The results of the decomposition appear in Table 5. In the table, we decompose the sources of change in earnings at the $90^{th}$, $50^{th}$, and $10^{th}$ percentiles of the earnings distribution as well as the sources of change in earnings inequality as measured by the 90-10 log earnings difference, the 90-50 log difference, and the 50-10 log difference. Panel (a) shows the decomposition in levels, while panel (b) provides the implied relative contribution of each component to the change in the statistic. In other words, panel (b) reports the change in the statistic when the factor is either assumed not to have occurred (as with worker and firm entry and exit) or replaced by its value in 1992 (as with observed worker characteristics and the conditional distribution of worker-firm matches).

The net changes in earnings levels and inequality reported in the final column, which echo those reported in Table 1, mask considerable flux in the earnings distribution due to changes in the underlying factors, which are spelled out in the intervening columns. First, an examination of column (2) reveals that sectoral changes in the distribution of employment led to small increases

in both the 90-50 and 50-10 log earnings differences, due largely to a reallocation of employment into industries with relatively low earnings (the $10^{th}$ and $50^{th}$ percentiles of earnings declined relative to the $90^{th}$). For instance, had there been no change in the sectoral distribution of employment, the 90-50 log earnings difference would have been 0.975 instead of the actual 0.981 (see panel (a)). Thus, changes in the sectoral distribution of employment contributed to an increase the 90-50 log earnings difference of 0.006 (see panel (b)). The marginal contribution of worker entry and exit conditional on changes in the sectoral distribution of employment had virtually no impact on the 50-10 log earnings difference and resulted in a slight decrease in the 90-50 log earnings difference.

More important were changes in observable worker characteristics, which led to decreases in both the 50-10 and 90-50 log earnings differences. These decreases came about as changes in observable worker characteristics, which in this case primarily reflect changes in average labour market experience, boosted earnings at the $10^{th}$ percentile more so than at the $50^{th}$ and $90^{th}$. The entry and exit of firms reinforced the effect of changes in worker characteristics, leading to sizable decreases in the 50-10 and 90-50 log earnings differences. Firm turnover had its largest effect at the lower end of the earnings distribution, propping up earnings at the $10^{th}$ percentile.

However, these effects were largely offset by the impact of sorting of workers and firms and the residual component, as columns (6) and (7) indicate. The reallocation of workers across firms clearly played a large role in determining changes in the shape of the overall earnings distribution, with changes in the joint distribution of worker skill and firm pay policies tending to work against individuals at the lower end of the earnings distribution and favoring those at the upper end. This sorting worked in tandem with the residual component to increase overall inequality. The residual effect could reflect changes in the returns to skills; in that case, the results are consistent with skill-biased technical change that favored those at the upper end of the earnings distribution and acted to increase the 90-50 log earnings difference more so than the 50-10 difference.

*6.2 Results by industry*

In Tables 6 and 7, we break out the results by industry. For the sake of brevity, we present only the implied changes in earnings levels and inequality driven by each factor. Table 6 presents

results for the four industries in which inequality (as measured by the 90-10 log earnings difference) decreased between 1992 and 2003 (agriculture, mining, construction, and retail trade). Table 7 presents results for the five industries in which inequality increased (manufacturing, TCU, wholesale trade, FIRE, and services).

Looking first at the four industries in which overall inequality declined, there are several striking similarities and differences in the factors driving changes in the distributions of earnings over time. Despite the high levels of worker churning across all industries, the second column of Table 6 suggests that the churning was among workers of the same average skill level ($\theta$), resulting in basically no change in inequality. That this is true in every industry suggests that, by and large, workforce quality within each industry is quite persistent, which is consistent with work by Haltiwanger et al. (2006). An analysis of the third column reveals that, holding $\theta$ constant, changes in the characteristics of workers (which largely reflects increases in experience) acted to decrease earnings inequality in three of the four industries that experienced declines in inequality between 1992 and 2003. This generally led to higher earnings at both ends of the distribution, but with a larger impact at the $10^{th}$ percentile than at the $90^{th}$ (except in retail), thus decreasing inequality.

The entry and exit of firms clearly had an enormous impact on the earnings distribution, as is evident from column (4) of Table 6. Holding workforce composition constant, if no firm entry or exit had occurred in the mining industry between 1992 and 2003, the 90-10 log earnings gap would have swung by 121 log points, most of which occurred between the $50^{th}$ and $10^{th}$ percentiles. Notably, across all industries, firm entry and exit typically acted to increase earnings at the bottom end of the distribution more than at the top, resulting in a decline in the 90-10 log earnings difference in each industry.

The effect of the sorting of workers and firms is evident in the fifth column of Table 6. Sorting of workers across different firms actually had a larger negative impact on earnings for workers in the $10^{th}$ percentile than in the $90^{th}$ percentile in each the four industries with declining overall inequality. Sorting acted to raise $90^{th}$ percentile earnings in three of the four industries. Still, for these four industries, the sorting effect was largely overshadowed by the effect of firm entry and exit.

The residual component was relatively small in each of the industries that experienced declining inequality, as the seventh column of Table 6 shows. Holding workforce and firm

composition as well as the allocation of workers across firms constant, other factors not included in the decomposition, which could include changes in the returns to observable and unobservable characteristics, worked to increase earnings inequality. It did so largely by depressing earnings at the lower end of the distribution relative to the upper end.

Table 7 reports the decompositions for the five industries in which overall earnings inequality (again measured as the 90-l0 log earnings gap) increased. As was the case for the declining-inequality industries, worker churning had little effect on the earnings distribution in these sectors. Meanwhile, the third column shows that changing observable worker characteristics lowered inequality by raising earnings more at the bottom than at the top of the earnings distribution in each of the industries. Interestingly, however, the magnitude of the impact of changing experience on earnings is quite different across industries. For example, changes in experience affected the $10^{th}$ and $50^{th}$ earnings percentiles in manufacturing by about 20 log points, compared with 11-13 log points in services.

The effect of firm entry and exit was substantial in these five industries. In general, turnover among businesses led to a decrease in earnings inequality by bolstering earnings at the bottom more than at the top end of the distribution. In wholesale trade, the $90^{th}$ percentile of earnings dropped considerably due to firm entry and exit.

The fifth column of Table 7 indicates that sorting of workers among firms generally led to an increase in inequality. Earnings at the bottom of the distribution were much lower due to sorting, leading to a rise in all three inequality measures in all industries. This effect was especially large in manufacturing and services. The contributions of the residual component followed a similar pattern, also tending to boost overall inequality in each of the five industries in Table 7. However, perhaps again reflecting skill-biased technical change that benefited workers at the upper end of the distribution more than at the lower end, the residual component tended to increase the 90-50 log earnings difference more so than the 50-10 difference.

Taken together, the decompositions in Table 6 and Table 7 show that, while trends in overall inequality as measured by the 90-10 log earnings difference diverged across industries, similar factors were at work beneath the surface. Worker entry and exit had little effect on earnings inequality measures despite high levels of worker churning in the economy. Changes in observable worker characteristics within each industry, which can largely be ascribed to changes in experience, acted to increase earnings at all levels, but tended to have a larger impact at the

lower end of the distribution. Firm entry and exit and the sorting of workers across firms had larger effects on earnings distributions across industries, with the former acting to decrease inequality and the latter acting to increase it in most industries. The residual component, which captures changes in returns to characteristics and other factors not accounted for in the decomposition, reinforced the effect of sorting, acting to further increase inequality. Nonetheless, despite the similarities in underlying factors, the size of these effects differed considerably across industries. Perhaps even more strikingly, even in industries in which there were small net changes in earnings distributions, there were often very large, offsetting effects from the underlying forces driving changes in inequality over time.

## 7. Minimum Wage Legislation

One of our central findings is the very large and offsetting effects of firm turnover and sorting among workers and firms. An examination of Tables 5-7 reveals that the largest manifestation of this phenomenon occurs in the 10th percentile of the distribution. This result raises the possibility that the adjustment is due to the substantial increase of the minimum wage in California. All four states in the sample had minimum wages of $4.25 per hour in 1992 and experienced incremental federally-mandated increases to $5.15 per hour by September 1997. However, California increased its minimum wage further thereafter, raising it to $5.75 by March 1998, $6.25 by January 2001, and $6.75 by January 2002. In the meantime, the minimum wage in Illinois, Maryland, and North Carolina remained at $5.15 per hour between 1997 and 2003. Given this, we would expect that if the minimum wage does have an effect on the earnings distribution, it would be apparent in the contribution of firm entry and exit on the earnings distribution for California in particular. The 1992 California firms included in the counterfactual would have been at a substantially lower minimum wage, and thus have lower estimated firm fixed effects, than California firms existing in 2003.

We examine the possible influence of the minimum wage legislation in California in Table 8, where we present results of the decomposition for retail trade for a sample that excludes California. Of all nine major industries, the retail trade industry employs the largest fraction of individuals working at or below the minimum wage (approximately 9% according to CPS data

for 2002).[6] In the retail trade industry, holding workforce composition constant, the counterfactual had there been no entry and exit of firms is that the log earnings at the 10th percentile of the distribution in services would increase 0.48 if California is included (see Table 6). Without California, the same counterfactual is that the log earnings would rise by only 0.91, as is evident in Table 8. At the 50th percentile, the effect of firm entry is about 0.13 whether California is included in the sample or not. This pattern runs counter to our expectation that the minimum wage hikes would have tended to increase the $\psi$ s of entering firms at the low end of the earnings distribution in California. Instead, it seems that firm entry actually had a larger positive effect on earnings at the low end of the distribution in Illinois, Maryland, and North Carolina, where there were no changes in the minimum wage, than in California, where there was.

Of course, other factors may also be at play. It is possible, for example, that the adjustment to changes in the minimum wage took place more on the employment and/or hours margin than the earnings margin. In that case, a muted response to the minimum wage increase in California might be due to exiting firms hiring relatively more part-time workers (who tend to be at the lower end of the earnings distribution) and entering firms hiring relatively more full-time workers (who tend to be higher in the earnings distribution). However, recent research suggests that increases in the minimum wage have little effect on hours worked (Zavodny, 2000; Sabia, 2008). Moreover, data from the CPS suggests little difference in either the share of employment in retail overall or trends in part-time employment in California versus the other three states in the sample. In California, as in Illinois, Maryland, and North Carolina together, the percentage of employment in retail trade remained roughly constant at 15% between 1992 and 2003. Meanwhile, part-time employment in retail trade fell from 31% in 1992 to 26% in 2003 in California, not markedly different than the decline from 34% to 27% observed in the other three states in the sample. While there may be other explanations for the limited evidence of any substantial impact of California's minimum wage legislation in the decomposition, we infer from these results that such institutional factors are not behind the large estimated effects we find, particularly with respect to the role of firm entry and exit as a source of changes in the earnings distribution over time.

[6] Under SIC codes, retail trade includes eating and drinking places, which employ a very large share of those individuals working at or below the minimum wage.

## 8. Conclusion

In this paper, we use linked employer-employee dataset from the Longitudinal Employer Household Dynamics Program at the U.S. Census Bureau to explore changes in the earnings distributions across sectors of the economy. We investigate how changes in workforce composition, firm entry and exit, and the matching of workers and firms affect economy-wide and industry-specific earnings distributions.

While there were differences across industries in the magnitudes and directions of change in various aspects of the earnings distribution between 1992 and 2003, our earnings decompositions suggest that most factors had similar qualitative effects across all industries. In particular, even in industries in which there was very little change in the aggregate earnings distribution between 1992 and 2003, there were enormous, albeit offsetting, changes in the factors contributing to changes in that distribution. Similar factors were at work in industries with declining inequality as well as those with increasing inequality. The magnitudes of these effects, however, varied considerably.

We find that worker entry and exit had very little impact on changes in earnings distributions over this time period for the industries examined. In other words, although worker turnover rates were high across industries, the average characteristics of industry workforces remained, by and large, very similar. Meanwhile, changes in observable characteristics among workers who stayed in an industry over the period, which largely reflect experience effects, tended to shift the earnings distributions of all industries to the right. In every industry but one, changes in observable worker characteristics also worked to decrease earnings inequality, in each case primarily by increasing earnings at the bottom of the earnings distribution relative to the top end.

The net impact of firm entry and exit was to reduce the dispersion of earnings overall. In nearly all industries, firm turnover acted to increase earnings at the bottom end of the distribution more than at the top. This effect persisted even after excluding from the sample one state that experienced a sizable increase in its minimum wage during the sample period, which might otherwise be expected to account for some of the large increases in earnings attributed to firm turnover at the lower end of the distribution in certain industries. While our results do not imply that changes in the minimum wage have no effect on the earnings distribution, they do suggest

that their effects do not manifest themselves through changes in the composition of firms in affected industries.

Meanwhile, sorting of workers and firms over time tended to increase the dispersion of industry earnings distributions between 1992 and 2003. Low wage workers are increasingly relegated to low wage firms, while high wage workers are increasingly finding their ways to high wage firms. This trend has worked to increase earnings inequality. Though the mechanism driving this sorting is unclear, it is consistent with skill-biased technical change to the extent that it might increase the relative returns for capital intensive firms to hiring highly skilled workers. Also consistent with skill-biased change is the substantive role of the residual component in the decomposition, which could reflect changes in the returns to observable and unobservable characteristics and act to increase inequality more at the upper end of the earnings distribution than the lower end.

Overall, using matched employee-employer panel data, we find that underlying even very small changes in earnings distributions over time are potentially very large but offsetting effects of firm turnover and the sorting of workers across firms. The extensive amounts of worker and firm reallocation in the U.S. economy, which Haltiwanger et al. (2006) and others have shown to have important implications for productivity, also clearly play a key role in shaping the distributions of earnings within and across industries. In particular, the entry and exit of firms and the sorting of workers across firms based on underlying worker skills are important determinants of changes in earnings inequality over time.

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**Table 1**
Earnings levels, differences, and changes by sector, 1992-2003.

| | 2003 Earnings Percentiles | | | 90-10 Log Wage Difference | | 90-50 Log Wage Difference | | 50-10 Log Wage Difference | |
|---|---|---|---|---|---|---|---|---|---|
| | 90th | 50th | 10th | 2003 | Change from 1992 | 2003 | Change from 1992 | 2003 | Change from 1992 |
| Agriculture | $44,149 | $19,234 | $9,126 | 1.58 | -0.14 | 0.83 | -0.02 | 0.75 | -0.12 |
| Mining | $82,705 | $45,879 | $22,427 | 1.30 | -0.09 | 0.59 | -0.06 | 0.72 | -0.03 |
| Construction | $73,174 | $34,181 | $14,831 | 1.60 | -0.12 | 0.76 | -0.01 | 0.83 | -0.11 |
| Manufacturing | $90,650 | $34,176 | $15,183 | 1.79 | 0.13 | 0.98 | 0.16 | 0.81 | -0.02 |
| TCU | $82,987 | $39,597 | $15,959 | 1.65 | 0.13 | 0.74 | 0.12 | 0.91 | 0.01 |
| Wholesale | $96,084 | $34,852 | $15,307 | 1.84 | 0.11 | 1.01 | 0.12 | 0.82 | 0.00 |
| Retail | $51,404 | $19,820 | $8,512 | 1.80 | -0.05 | 0.95 | 0.01 | 0.85 | -0.06 |
| FIRE | $114,428 | $37,083 | $16,244 | 1.95 | 0.18 | 1.13 | 0.15 | 0.83 | 0.03 |
| Services | $83,079 | $31,346 | $11,523 | 1.98 | 0.05 | 0.97 | 0.08 | 1.00 | -0.03 |
| All Sectors | $82,207 | $31,477 | $11,992 | 1.93 | 0.06 | 0.96 | 0.10 | 0.97 | -0.03 |

Note: Based on LEHD data from CA, IL, MD and NC.

**Table 2**
Worker mobility in and out of industrial sectors, 1992-2003.

| | | Proportion in Industry Sector | | |
|---|---|---|---|---|
| | Number of Workers in 1992 and 2003 | Only in 1992 | Only in 2003 | Both Years |
| Agriculture | 578,036 | 39% | 48% | 13% |
| Mining | 67,888 | 56% | 29% | 14% |
| Construction | 1,511,595 | 32% | 53% | 14% |
| Manufacturing | 5,145,894 | 44% | 35% | 21% |
| TCU | 1,775,581 | 37% | 44% | 19% |
| Wholesale Trade | 2,006,918 | 41% | 47% | 12% |
| Retail Trade | 4,214,151 | 39% | 49% | 12% |
| FIRE | 2,101,998 | 36% | 47% | 17% |
| Services | 10,196,180 | 31% | 51% | 18% |

Note: Based on LEHD data for CA, IL, MD and NC. Some rows may not sum to 100% due to rounding.

**Table 3**
Changes in workforce composition between 1992 and 2003.

| | | Change in | | | | | |
|---|---|---|---|---|---|---|---|
| | Employment in 1992 | Employment between 1992 and 2003 | Proportion of Workforce | | | | Individual Fixed Effects (log points) |
| | | | Male | 14-29 | 30-49 | 50+ | |
| Agriculture | 300,709 | 17% | -7% | -6% | 1% | 5% | 0.05 |
| Mining | 48,063 | -39% | 2% | 0% | -11% | 10% | 0.08 |
| Construction | 704,268 | 46% | -1% | -5% | 1% | 4% | 0.03 |
| Manufacturing | 3,357,441 | -14% | 2% | -7% | -1% | 7% | 0.06 |
| TCU | 991,212 | 14% | 2% | -3% | -4% | 7% | 0.06 |
| Wholesale Trade | 1,066,376 | 11% | 0% | -7% | 0% | 7% | 0.06 |
| Retail Trade | 2,138,239 | 20% | 0% | -5% | 1% | 4% | 0.04 |
| FIRE | 1,111,889 | 21% | 3% | -5% | -1% | 6% | 0.05 |
| Services | 4,998,570 | 41% | 0% | -4% | -4% | 7% | 0.06 |

Note: Based on LEHD data for CA, IL, MD and NC.

**Table 4**
Firm entry and exit rates.

| | Number of Firms in 1992 and 2003 | Proportion in Industry Sector | | |
|---|---|---|---|---|
| | | Only in 1992 | Only in 2003 | In Both Years |
| Agriculture | 50,825 | 32% | 39% | 29% |
| Mining | 2,135 | 45% | 35% | 20% |
| Construction | 155,195 | 33% | 45% | 22% |
| Manufacturing | 107,200 | 36% | 37% | 27% |
| TCU | 56,355 | 35% | 45% | 20% |
| Wholesale Trade | 143,414 | 36% | 43% | 21% |
| Retail Trade | 263,093 | 40% | 41% | 20% |
| FIRE | 120,763 | 33% | 46% | 22% |
| Services | 686,606 | 31% | 49% | 19% |

Note: Based on LEHD data for CA, IL, MD and NC. Some rows may not sum to 100% due to rounding.

**Table 5**
Decompositions of changes in log earnings and earnings inequality measures, 1992-2003: All sectors.

| | (a) Levels | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2003 | Sector Distribution | +Worker entry and exit | + Change in observable worker characteristics | + Firm entry and exit | +Sorting of firms and workers | 1992 | Change from 1992 to 2003 |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| 10th percentile | 9.379 | 9.393 | 9.403 | 9.245 | 8.261 | 9.082 | 9.276 | 0.103 |
| 50th percentile | 10.341 | 10.347 | 10.357 | 10.209 | 10.059 | 10.188 | 10.271 | 0.070 |
| 90th percentile | 11.322 | 11.322 | 11.337 | 11.254 | 11.299 | 11.242 | 11.155 | 0.167 |
| 90-10 difference | 1.943 | 1.929 | 1.934 | 2.009 | 3.038 | 2.160 | 1.879 | 0.064 |
| 90-50 difference | 0.981 | 0.975 | 0.980 | 1.045 | 1.240 | 1.054 | 0.884 | 0.097 |
| 50-10 difference | 0.962 | 0.954 | 0.954 | 0.964 | 1.798 | 1.106 | 0.995 | -0.033 |

| | (b) Change in Statistic when also Accounting for | | | | | |
|---|---|---|---|---|---|---|
| | Sector Distribution | Worker entry and exit | Change in observable worker characteristics | Firm entry and exit | Sorting of firms and workers | Residual |
| 10th percentile | -0.014 | -0.010 | 0.158 | 0.984 | -0.821 | -0.194 |
| 50th percentile | -0.006 | -0.010 | 0.148 | 0.150 | -0.129 | -0.083 |
| 90th percentile | 0.000 | -0.015 | 0.083 | -0.045 | 0.057 | 0.087 |
| 90-10 difference | 0.014 | -0.005 | -0.075 | -1.029 | 0.878 | 0.281 |
| 90-50 difference | 0.006 | -0.005 | -0.065 | -0.195 | 0.186 | 0.170 |
| 50-10 difference | 0.008 | 0.000 | -0.010 | -0.834 | 0.692 | 0.111 |

Note: Based on LEHD data for CA, IL, MD and NC.

**Table 6**
Decompositions of changes in log earnings and earnings inequality measures, 1992-2003: Industries with declining inequality.

| | | Change in Statistic when also Accounting for | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2003 | Worker entry and exit | Change in observable worker char. | Firm entry and exit | Sorting of firms and workers | Residual | 1992 | Change from 1992 to 2003 |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| **Agriculture** | | | | | | | | |
| 10th percentile | 9.12 | -0.01 | 0.21 | 0.26 | -0.20 | 0.01 | 8.84 | 0.28 |
| 50th percentile | 9.86 | -0.01 | 0.14 | 0.04 | -0.02 | 0.01 | 9.70 | 0.16 |
| 90th percentile | 10.70 | -0.01 | 0.12 | -0.11 | 0.10 | 0.04 | 10.56 | 0.14 |
| 90-10 difference | 1.58 | 0.00 | -0.08 | -0.37 | 0.29 | 0.02 | 1.72 | -0.14 |
| 90-50 difference | 0.83 | -0.01 | -0.02 | -0.15 | 0.12 | 0.03 | 0.85 | -0.02 |
| 50-10 difference | 0.75 | 0.00 | -0.07 | -0.23 | 0.18 | 0.00 | 0.87 | -0.12 |
| **Mining** | | | | | | | | |
| 10th percentile | 10.02 | 0.01 | 0.22 | 1.34 | -1.37 | -0.17 | 9.99 | 0.03 |
| 50th percentile | 10.73 | 0.01 | 0.11 | 0.29 | -0.36 | -0.06 | 10.74 | -0.01 |
| 90th percentile | 11.32 | 0.01 | -0.03 | 0.13 | -0.03 | -0.15 | 11.39 | -0.07 |
| 90-10 difference | 1.31 | -0.01 | -0.25 | -1.21 | 1.35 | 0.03 | 1.40 | -0.09 |
| 90-50 difference | 0.59 | 0.00 | -0.14 | -0.17 | 0.33 | -0.09 | 0.65 | -0.06 |
| 50-10 difference | 0.72 | 0.00 | -0.12 | -1.04 | 1.01 | 0.12 | 0.75 | -0.03 |
| **Construction** | | | | | | | | |
| 10th percentile | 9.61 | -0.01 | 0.16 | 0.56 | -0.36 | -0.15 | 9.41 | 0.20 |
| 50th percentile | 10.44 | -0.02 | 0.12 | 0.16 | -0.08 | -0.09 | 10.35 | 0.09 |
| 90th percentile | 11.20 | -0.02 | 0.07 | 0.03 | 0.01 | 0.00 | 11.13 | 0.08 |
| 90-10 difference | 1.60 | -0.01 | -0.09 | -0.53 | 0.37 | 0.14 | 1.72 | -0.12 |
| 90-50 difference | 0.76 | 0.00 | -0.06 | -0.12 | 0.09 | 0.09 | 0.77 | -0.01 |
| 50-10 difference | 0.84 | -0.01 | -0.04 | -0.41 | 0.28 | 0.06 | 0.94 | -0.11 |
| **Retail Trade** | | | | | | | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 10th percentile | 9.05 | -0.01 | 0.11 | 0.48 | -0.36 | -0.10 | 8.93 | 0.12 |
| 50th percentile | 9.89 | -0.01 | 0.14 | 0.13 | -0.06 | -0.14 | 9.83 | 0.06 |
| 90th percentile | 10.85 | -0.01 | 0.12 | -0.02 | 0.07 | -0.09 | 10.78 | 0.07 |
| 90-10 difference | 1.80 | -0.01 | 0.01 | -0.50 | 0.43 | 0.01 | 1.85 | -0.05 |
| 90-50 difference | 0.95 | 0.00 | -0.02 | -0.15 | 0.13 | 0.05 | 0.94 | 0.01 |
| 50-10 difference | 0.85 | 0.00 | 0.03 | -0.35 | 0.30 | -0.04 | 0.90 | -0.06 |

Note: Based on LEHD data for CA, IL, MD and NC.

**Table 7**
Decompositions of changes in log earnings and earnings inequality measures, 1992-2003: Sectors with rising inquality.

| | | Change in Statistic when also Accounting for | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2003 | Worker entry and exit | Change in observable worker characteristics | Firm entry and exit | Sorting of firms and workers | Residual | 1992 | Change from 1992 to 2003 |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Manufacturing | | | | | | | | |
| 10th percentile | 9.63 | 0.01 | 0.20 | 1.35 | -1.29 | -0.18 | 9.55 | 0.08 |
| 50th percentile | 10.44 | 0.01 | 0.21 | 0.13 | -0.25 | -0.04 | 10.38 | 0.06 |
| 90th percentile | 11.42 | 0.00 | 0.13 | -0.08 | -0.01 | 0.17 | 11.20 | 0.22 |
| 90-10 difference | 1.79 | 0.00 | -0.07 | -1.43 | 1.28 | 0.35 | 1.65 | 0.14 |
| 90-50 difference | 0.98 | 0.00 | -0.08 | -0.21 | 0.24 | 0.21 | 0.82 | 0.16 |
| 50-10 difference | 0.81 | 0.00 | 0.01 | -1.22 | 1.05 | 0.13 | 0.83 | -0.02 |
| TCU | | | | | | | | |
| 10th percentile | 9.68 | -0.01 | 0.17 | 0.70 | -0.69 | -0.20 | 9.70 | -0.02 |
| 50th percentile | 10.59 | -0.01 | 0.09 | 0.13 | -0.17 | -0.06 | 10.61 | -0.02 |
| 90th percentile | 11.33 | -0.01 | 0.07 | 0.01 | -0.05 | 0.08 | 11.22 | 0.11 |
| 90-10 difference | 1.65 | 0.01 | -0.11 | -0.69 | 0.65 | 0.28 | 1.52 | 0.13 |
| 90-50 difference | 0.74 | 0.01 | -0.03 | -0.12 | 0.13 | 0.14 | 0.62 | 0.13 |
| 50-10 difference | 0.91 | 0.00 | -0.08 | -0.57 | 0.52 | 0.14 | 0.90 | 0.01 |
| Wholesale Trade | | | | | | | | |
| 10th percentile | 9.64 | 0.00 | 0.17 | 0.46 | -0.38 | -0.17 | 9.55 | 0.09 |
| 50th percentile | 10.46 | 0.00 | 0.15 | 0.02 | -0.09 | 0.01 | 10.38 | 0.08 |
| 90th percentile | 11.47 | 0.00 | 0.08 | -0.38 | 0.29 | 0.21 | 11.28 | 0.20 |
| 90-10 difference | 1.84 | 0.00 | -0.10 | -0.84 | 0.67 | 0.38 | 1.73 | 0.11 |
| 90-50 difference | 1.01 | 0.00 | -0.07 | -0.40 | 0.38 | 0.20 | 0.90 | 0.12 |
| 50-10 difference | 0.82 | 0.00 | -0.03 | -0.44 | 0.29 | 0.17 | 0.83 | 0.00 |
| FIRE | | | | | | | | |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 10th percentile | 9.70 | -0.01 | 0.18 | 0.79 | -0.66 | -0.13 | 9.53 | 0.17 |
| 50th percentile | 10.52 | -0.01 | 0.16 | 0.14 | -0.08 | -0.02 | 10.32 | 0.20 |
| 90th percentile | 11.65 | -0.01 | 0.09 | -0.10 | 0.20 | 0.16 | 11.30 | 0.35 |
| 90-10 difference | 1.95 | 0.00 | -0.09 | -0.89 | 0.86 | 0.30 | 1.77 | 0.18 |
| 90-50 difference | 1.13 | 0.00 | -0.07 | -0.24 | 0.28 | 0.18 | 0.97 | 0.15 |
| 50-10 difference | 0.83 | 0.00 | -0.02 | -0.65 | 0.58 | 0.11 | 0.80 | 0.03 |
| Services | | | | | | | | |
| 10th percentile | 9.35 | 0.02 | 0.13 | 1.27 | -0.99 | -0.32 | 9.24 | 0.11 |
| 50th percentile | 10.35 | 0.02 | 0.11 | 0.19 | -0.08 | -0.16 | 10.27 | 0.08 |
| 90th percentile | 11.33 | 0.01 | 0.04 | 0.00 | 0.11 | 0.00 | 11.16 | 0.17 |
| 90-10 difference | 1.98 | -0.01 | -0.09 | -1.26 | 1.10 | 0.32 | 1.92 | 0.05 |
| 90-50 difference | 0.98 | -0.01 | -0.07 | -0.18 | 0.18 | 0.16 | 0.89 | 0.08 |
| 50-10 difference | 1.00 | 0.00 | -0.03 | -1.08 | 0.92 | 0.16 | 1.03 | -0.03 |

Note: Based on LEHD data for CA, IL, MD and NC.

**Table 8**
Decompositions of changes in log earnings and earnings inequality measures, 1992-2003: Excluding California

| | | Change in Statistic when also Accounting for | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2003 (1) | Worker entry and exit (2) | Change in observable worker characteristics (3) | Firm entry and exit (4) | Sorting of firms and workers (5) | Residual (6) | 1992 (7) | Change from 1992 to 2003 (8) |
| | | | | Retail Trade | | | | |
| 10th percentile | 9.72 | -0.01 | 0.18 | 0.91 | -0.78 | -0.13 | 9.54 | 0.18 |
| 50th percentile | 10.50 | -0.01 | 0.18 | 0.13 | -0.10 | 0.02 | 10.27 | 0.23 |
| 90th percentile | 11.62 | -0.01 | 0.11 | -0.17 | 0.23 | 0.21 | 11.26 | 0.36 |
| 90-10 difference | 1.91 | 0.00 | -0.08 | -1.08 | 1.01 | 0.33 | 1.72 | 0.18 |
| 90-50 difference | 1.12 | 0.00 | -0.08 | -0.30 | 0.33 | 0.18 | 0.99 | 0.14 |
| 50-10 difference | 0.78 | 0.00 | 0.00 | -0.78 | 0.68 | 0.15 | 0.73 | 0.05 |

Note: Based on LEHD data for IL, MD and NC.

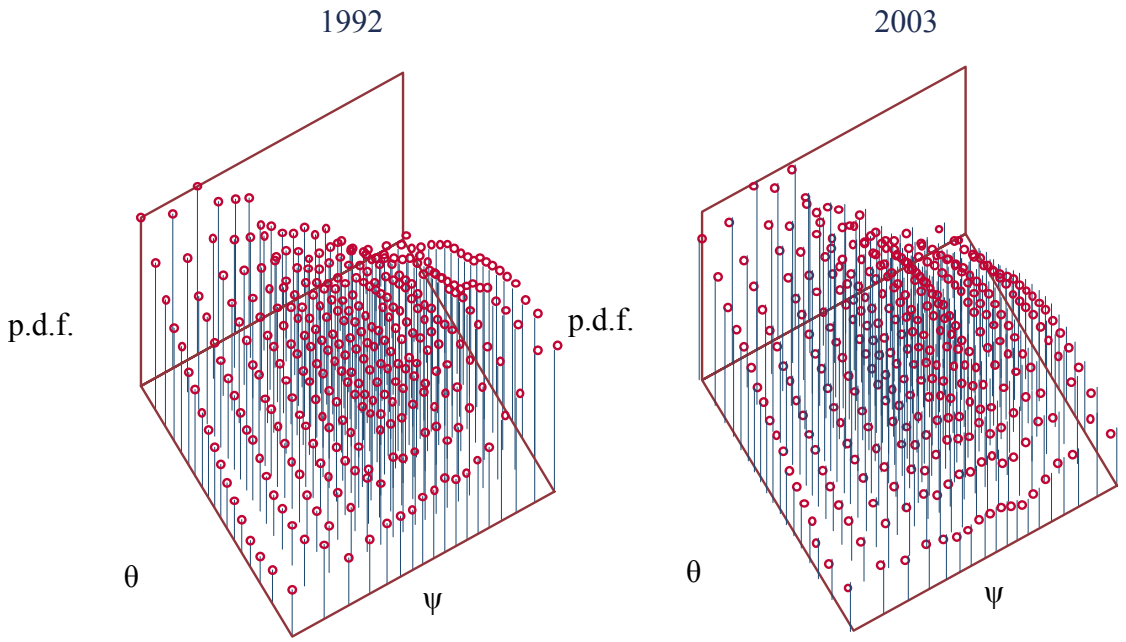



**Fig. 1.** Joint distribution of worker human capital ($\theta$) and firm pay policy ($\psi$) match.
Note: Based on LEHD data from CA, IL, MD and NC.